

ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

LECTURE 6

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CONTENTS:

- Communication of Enterprise Architectures
- A Language for Enterprise Modelling

COMMUNICATION OF ENTERPRISE ARCHITECTURES

Describing architectures is all about communication. The role of an architecture description is, it always involves some communicative aspect. Consider, as an illustration, the potential uses of architecture descriptions as identified in the IEEE 1471 standard (IEEE Computer Society 2000):

- Expression of the system and its (potential) evolution.
- Analysis of alternative architectures.
- Business planning for transition from a legacy architecture to a new architecture.
- Communications among organizations involved in the development, production, fielding, operation, and maintenance of a system.
- Communications between acquirers and developers as a part of contract negotiations.
- Criteria for certifying conformance of implementations to the architecture.
- Development and maintenance documentation, including material for reuse repositories and training material.
- Input to subsequent system design and development activities.
- Input to system generation and analysis tools.
- Operational and infrastructure support; configuration management and repair; redesign and maintenance of systems, subsystems, and components.
- Planning and budget support.
- Preparation of acquisition documents (e.g., requests for proposal and statements of work).
- Review, analysis, and evaluation of the system across the life cycle.
- Specification for a group of systems sharing a common set of features, (e.g., product lines).

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System Development as a Knowledge Transformation Process

In essence, we regard system development as a knowledge transformation process whereby conversations are used to share and create knowledge pertaining to both the system being developed, as well as the development process itself. The notion of '**conversation**' should be interpreted here in the broadest sense, ranging from a single person producing an (architectural) description, via a one-on-one design or elicitation session, to a workshop with several stakeholders, and even the widespread dissemination of definitive architectures.

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System Development as a Knowledge Transformation Process

- **System development community**: a group of objects, such as actors and representations, which are involved in the development of a system.
- **Concern**: an interest of a stakeholder with regards to the architecture description of some system, resulting from the stakeholder's goals, and the present or future role(s) played by the system in relation to these goals.

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System Development as a Knowledge Transformation Process

Some examples of **concerns** are:

- The current situation with regards to the computerized support of a business process.
- The requirements of a specific stakeholder with regards to the desired situation.
- The improvements, which a new system may bring, to a pre-existing situation in relation to the costs of acquiring the system.
- The potential impact of a new system on the activities of a prospective user.
- The potential impact of a new system on the work of the system administrators that are to maintain the new system.

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System Development as a Knowledge Transformation Process

Once the knowledge has been introduced into a community, it can be shared among members of that community. **Sharing knowledge** between different actors may progress through a number of stages. We distinguish three major stages:

- **Aware:** An actor may become aware of (possible) knowledge by way of the sharing by another actor.
- **Agreed:** Once knowledge is shared, an actor can make up his or her own mind about it and decide whether or not to agree to the knowledge shared.
- **Committed:** Actors who agree to a specific knowledge topic may decide actually to commit to this knowledge. In other words, they may decide to adapt their future behavior in accordance with this knowledge.

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Conversation Strategies

Conversations take place in some situation that may limit the execution of the conversation. We may characterize such a situation further in terms of situational factors:

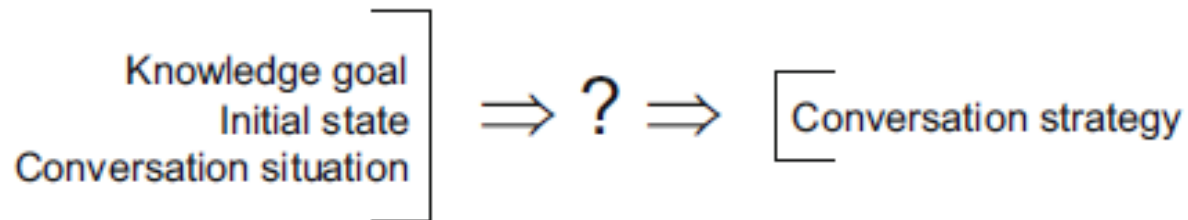
- **Availability of resources:** Refers to the availability of resources that can be used in a conversation. The availability of resources can be refined to more specific factors such as time for execution, actors, intellectual capacities needed from the actors, or financial means.
- **Complexity:** The resources needed for the conversation, the knowledge being conversed about, etc., will exhibit a certain level of complexity. This complexity also influences the conversation strategy to be followed. Examples of such complexity factors, are the heterogeneity of the actors involved, the quantity of actors, complexity of the technology used, the complexity of the knowledge being conversed about, and the size of the gap between the initial knowledge state and the desired knowledge state.
- **Uncertainty:** If you want to determine a conversation strategy fit for a given situation, you have to make assumptions about the knowledge goal.

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Conversation Strategies

According to Fig. 4.2. Given a knowledge goal, an initial state, and conversation situation, a conversation strategy can be determined, which should lead us from the initial state to the knowledge state as desired by the knowledge goal, taking into account the conversation situation at hand.

Fig. 4.2 From knowledge goal to conversation strategy



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Architectural Conversations

We can identify the following classes of knowledge goals that you may want to achieve in a conversation:

- **Introduction of knowledge:** This refers to situations where there is a need to introduce into or create new knowledge in a (part of a) development community. These kinds of knowledge goals typically lead to training or awareness sessions.
- **Agreement to knowledge:** With this class of knowledge goals, we refer to situations in which the mutual agreement of different stakeholders (with their own specific stakes and concerns!) needs to be improved or validated.
- **Commitment to knowledge:** In these cases, the knowledge goal goes beyond that of achieving agreement. Stakeholders should be willing to act upon the knowledge they agree to.

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Architectural Conversations: Conversation Techniques

- **Brown-paper session:** Structured brainstorm-like group session (up to about 15 people) in which items (keywords or short phrases) are elicited from the individuals in the group in answer to a question such as: 'What are the key functionality issues in our current IT architecture?'. The items are then collected on a sheet of paper and, by means of an open and creative group process, structured and categorized. This may involve adding, deleting, merging, or changing items.
- **Elicitation interview:** An interview where an analyst puts informative questions to the informants. The aim is to gather knowledge from the informants. Interviews can be more or less 'open': they can be strictly focused or guided, but the conversation can also be left open to go where the interest of the interviewer or informants leads it.
- **Workshop:** Involves one to, say, 15 people, working on a model or view interactively, mediated by an architect or analyst. This class also encompasses so-called joint modelling sessions.

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Architectural Conversations: Conversation Techniques

- **Validation interview:** An interview where an analyst will aim to find out if the view or model matches the views and expectations of an informant. This could be a view or model that has been communicated to the informants beforehand, or during the interview. A validation interview will typically be much more 'closed' than an elicitation interview: there will have to be some systematic approach by which validity of the view or model is checked.
- **Committing review:** A group of stakeholders are presented with a number of alternative models or views and their impact. They are requested to select one alternative and commit to this alternative based on their insights into the potential impact. This typically involves a formal decision-making processes.
- **Presentation:** Involves one to three people presenting a model or view to a group of, say, up to a hundred people. One may decide to elicit feedback, but this is usually gathered afterwards, in a more personal way, or at least 1:1 (e.g. through a round of feedback).
- **Mailing:** A form of 'mass' communication, where a model or view is presented or handed over to a large number of people. Feedback may or may not be encouraged.

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Architectural Conversations

Table 4.1 Knowledge goals and conversation techniques

<i>Conversation Technique</i>	<i>Knowledge Goal</i>		
	Introduce	Agree	Commit
Brown-paper session	++	+	—
Elicitation interview	++	+	—
Workshop	+	++	+
Validation interview	—	++	+
Committing review	—	—	++
Presentation	++	—	—
Mailing	+	—	—

A + indicates that a certain conversation class is well suited for the selected technique of knowledge goals, while ++ indicates that it is particularly well suited. On the other hand, a -- indicates that a certain conversation technique is ill-suited for the selected class of knowledge goals

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Discussion

According to the below paper choose four identified theories related to the key elements of an EA practice which more focus on communication and try to describe them?

- Kotusev, S., & Kurnia, S. (2021). *The theoretical basis of enterprise architecture: A critical review and taxonomy of relevant theories*. *Journal of Information Technology*, 36(3), 275-315.



A LANGUAGE FOR ENTERPRISE MODELLING

Architecture provides a means to handle the complexity of modern information intensive enterprises. To this end, architects need ways to express architectures as clearly as possible: both for **their own understanding** and for **communication with other stakeholders**, such as system developers, end users, and managers. However,...

- the current situation is that architects in different domains, even within the same organization, often use their own description techniques and conventions.
- there is no standard language for describing enterprise architectures in a precise way across domain borders.
- they are often described either in informal pictures that lack a well-defined meaning, or in detailed design languages (such as UML) that are difficult to understand for non-experts.

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In Fig. 5.1, the role that the enterprise architecture modelling language plays in our approach is summarized. It provides a means for integration, by allowing the creation of models that show high-level structures within domains and the relations between domains.

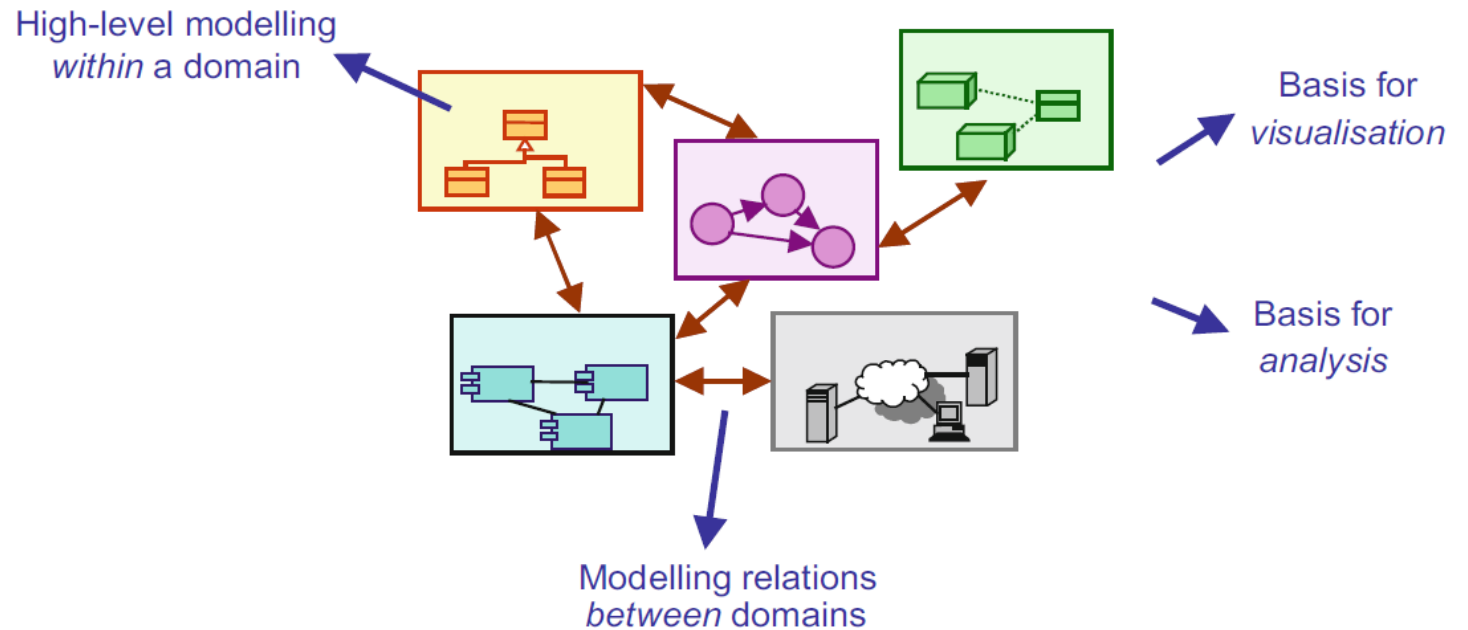


Fig. 5.1 | The role of the ArchiMate language

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Three main layers:

1. The **business layer** offers products and services to external customers, which are realized in the organization by business processes (performed by business actors or roles).
2. The **application layer** supports the business layer with application services which are realized by (software) application components.
3. The **technology layer** offers infrastructural services (e.g., processing, storage, and communication services) needed to run applications, realized by computer and communication devices and system software.

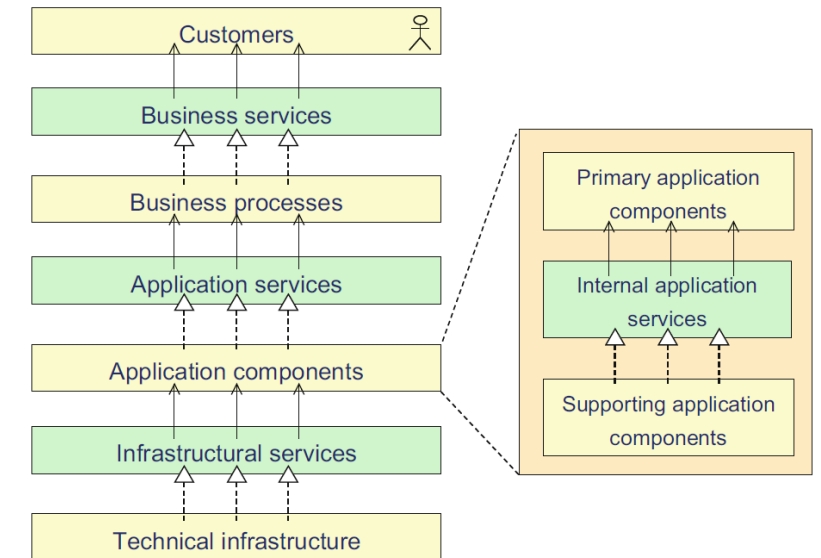


Fig. 5.2 Layered view

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A premise of the **ArchiMate language** is that the general structure of models within the different layers is similar. The same types of concepts and relations are used, although their exact nature and granularity differ. As a result of this uniformity, models created for the different layers can quite easily be aligned with each other. To identify the concepts that make up this general structure, we start from the three dimensions shown in Fig. 5.3.

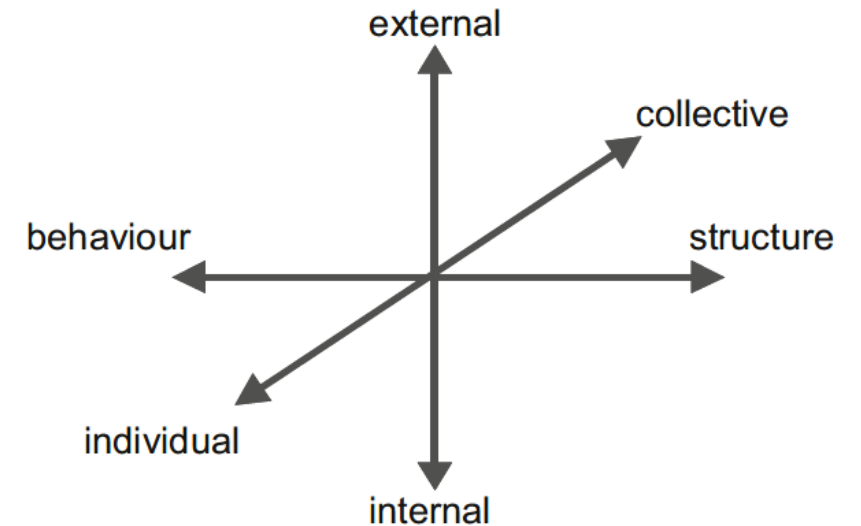


Fig. 5.3 Three dimensions of architectural concepts

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The core concepts that are found in each layer of the language are depicted in Fig. 5.4.

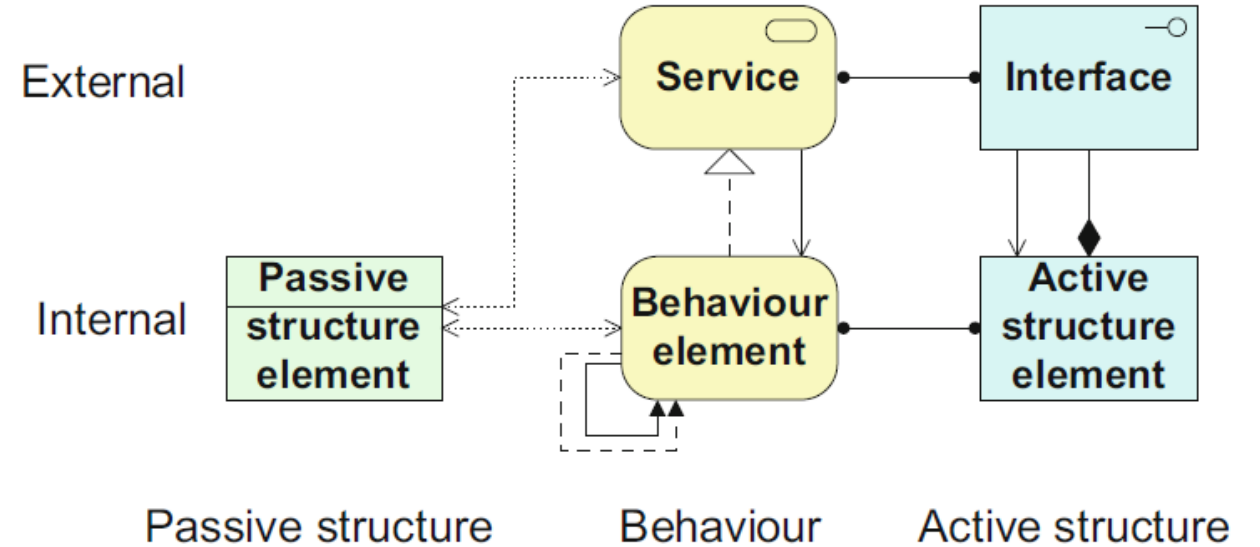


Fig. 5.4 The core concepts of the ArchiMate language

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The structure of the language and the most important concepts are summarized in Fig. 5.5.

	Passive structure	Behavior	Active structure	Motivation
Strategy	resources	courses of action, capabilities	resources	stakeholders, drivers, goals, principles and requirements
Business	business objects	business services, functions and processes	business actors and roles	
Application	data objects	application services, functions and processes	application components and interfaces	
Technology	artifacts	technology services, functions and processes	devices, system software, communication networks	
Physical	material		facilities, equipment, distribution networks	
Implementation & migration	deliverables	work packages	plateaus	

Fig. 5.5 Main concepts of the ArchiMate language

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A full overview of the language meta-model can be found in the official specification document (The Open Group 2016a) and the notation is given in Appendix A

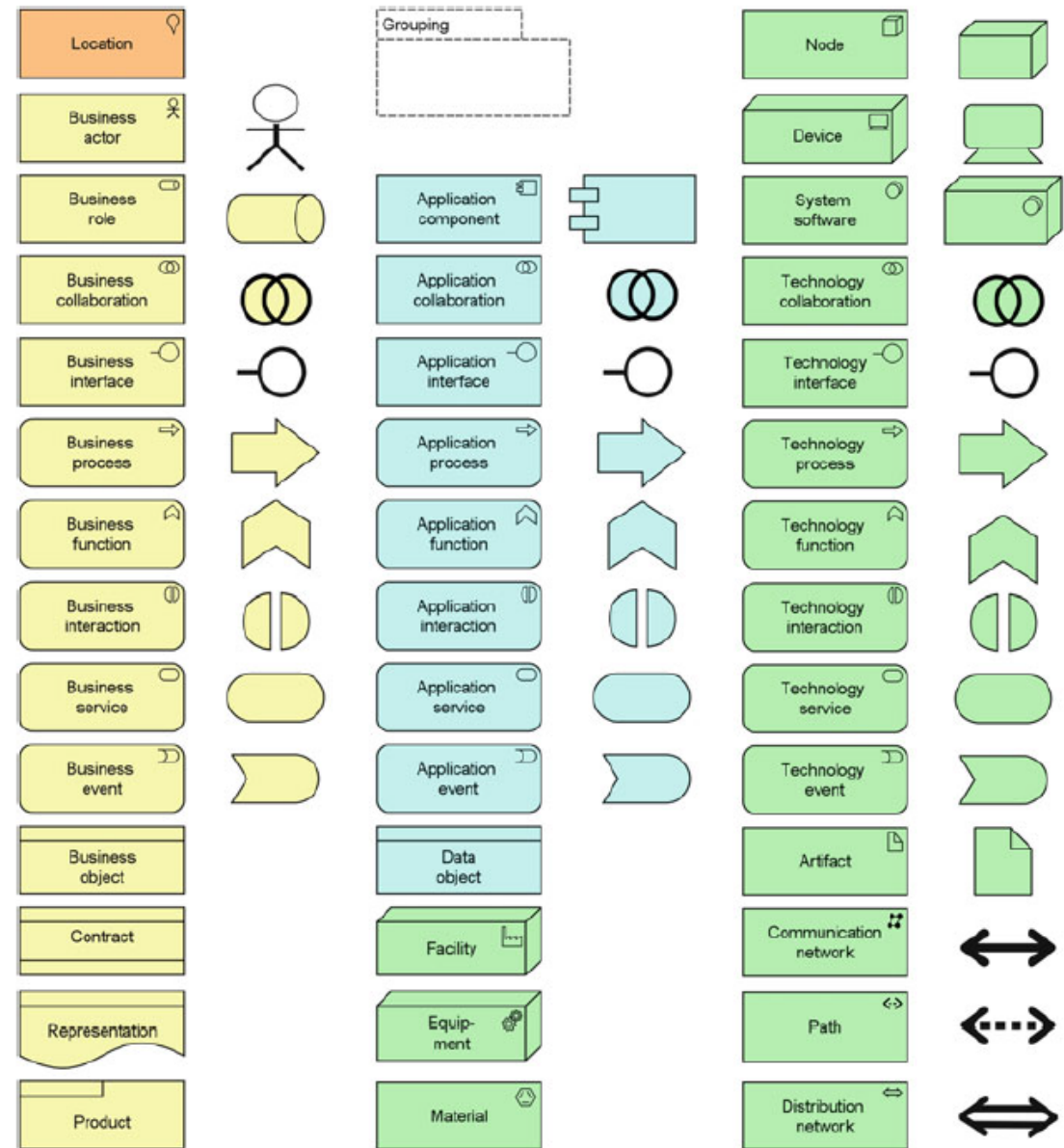


Fig. A.1 Symbols of the ArchiMate Core language

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An example of a motivation model is shown in Fig. 5.6. The example illustrates the use of the central concepts and relations.

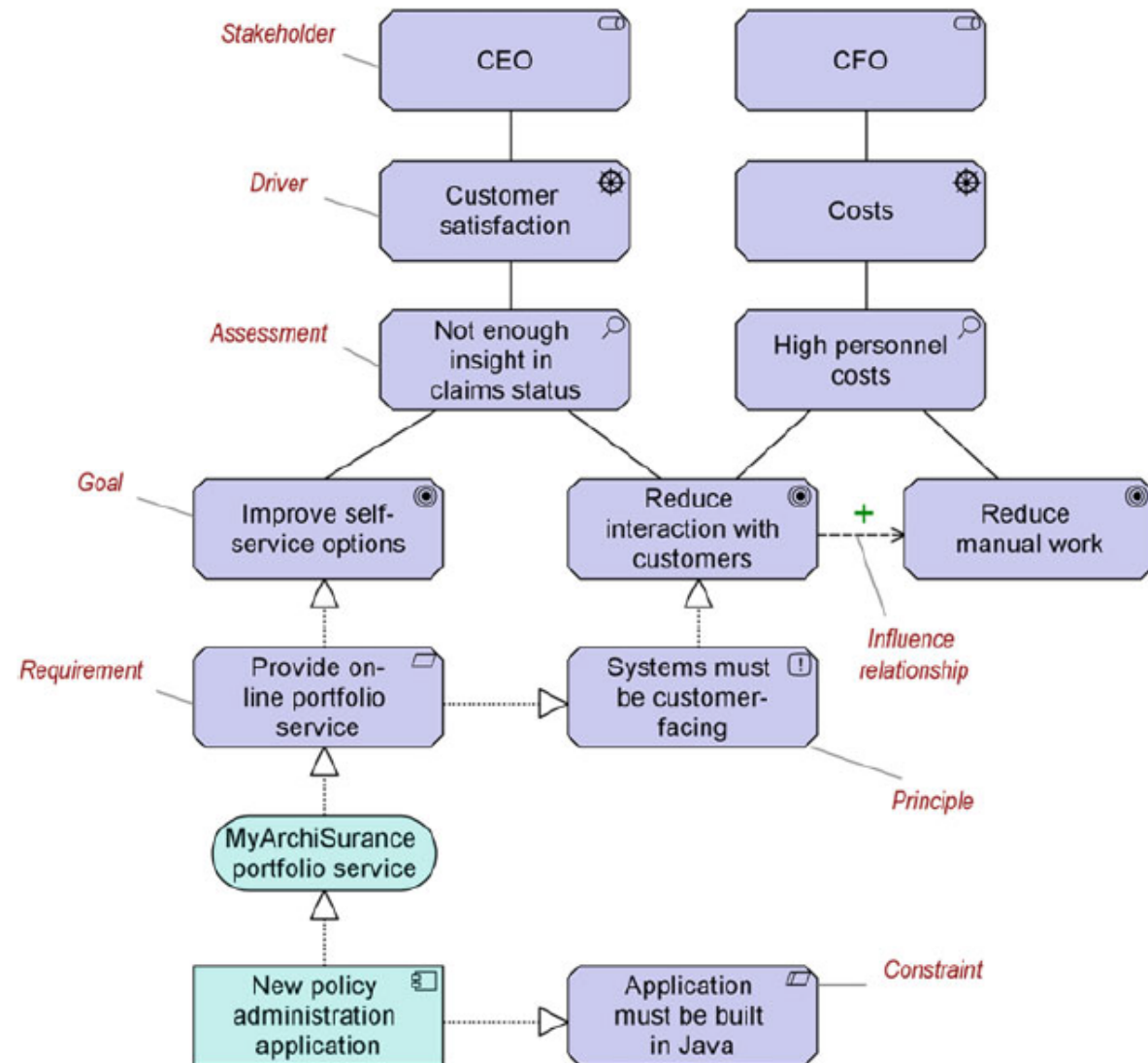


Fig. 5.6 Example of a motivation model

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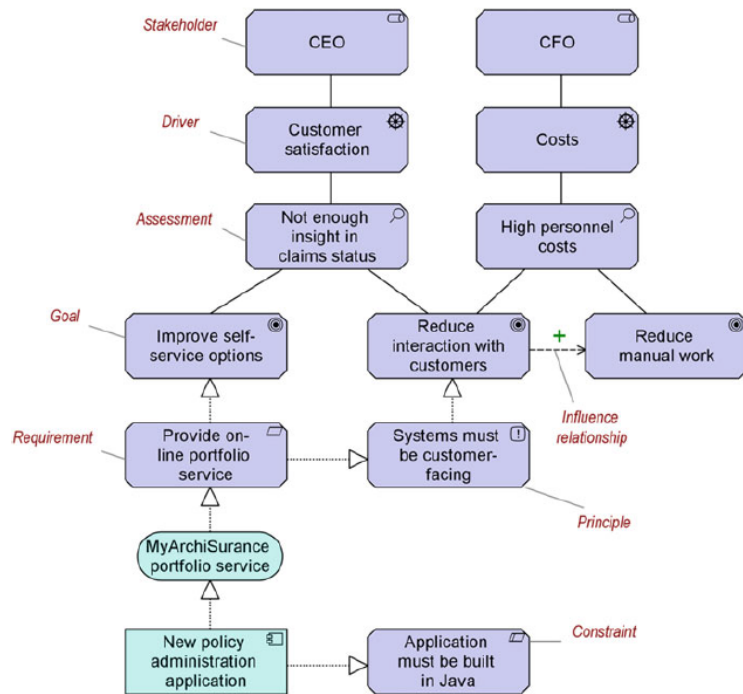


Fig. 5.6 Example of a motivation model

Motivation Concepts

- **Stakeholder:** the role of an individual, team or organization (or classes thereof) that represents their interest in the outcome of the architecture.
- **Driver:** an external or internal condition that motivates an organization to define its goals and implement the changes necessary to achieve them.
- **Assessment:** the result of an analysis of the state of affairs of the enterprise with respect to some driver.

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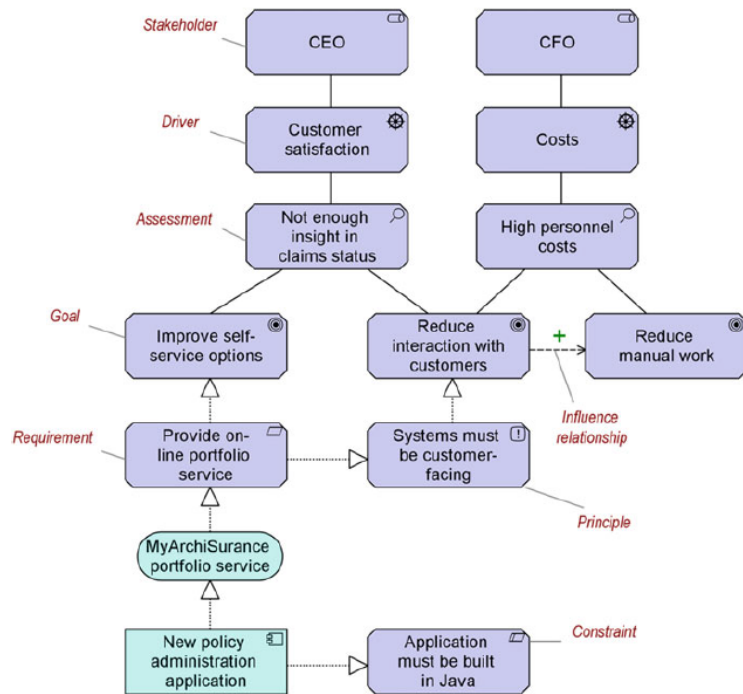


Fig. 5.6 Example of a motivation model

Motivation Concepts

- **Goal:** a high-level statement of intent, direction or desired end state for an organisation and its stakeholders.
- **Requirement:** a statement of need that must be met by the architecture.
- **Constraint:** a factor that prevents or obstructs the realisation of goals.
- **Principle:** a qualitative statement of intent that should be met by the architecture.
- **Outcome:** an end result that has been achieved.
- **Value:** the relative worth, utility or importance of a core element or an outcome.
- **Meaning:** the knowledge or expertise present in a business object or its representation, given a particular context.

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A course of action represents what an enterprise has decided to do to realize its goals. It can be used to model a high-level strategic plan or direction, or more concrete lower-level tactics. Courses of action are typically realized by capabilities and resources, as shown in Fig. 5.7.

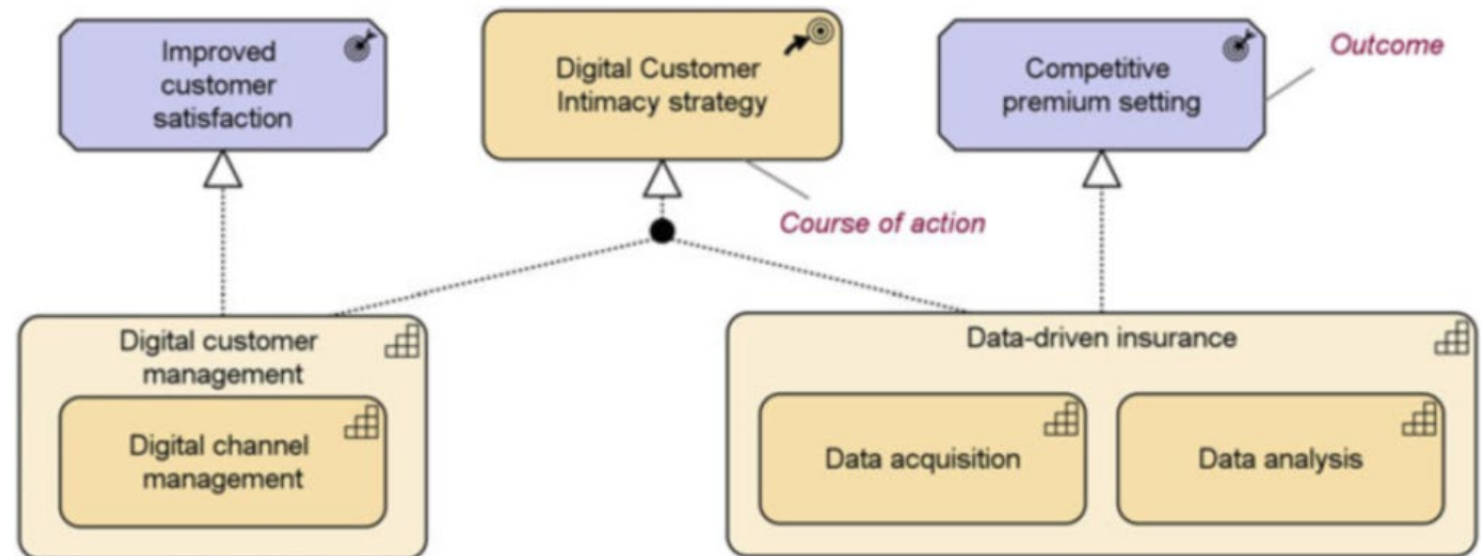


Fig. 5.7 Course of action and outcomes realised by capabilities

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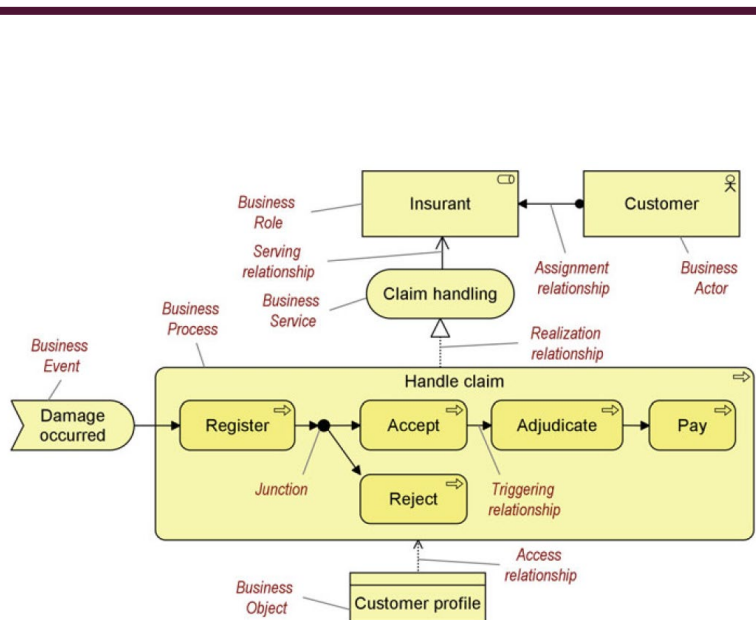


Fig. 5.9 Example of a business layer model

Business Structure Concepts

- **Business actor:** a business entity that is capable of performing behaviour.
- **Business role:** the responsibility for performing specific behaviour, to which an actor can be assigned, or the part an actor plays in a particular action or event.
- **Business collaboration:** an aggregate of two or more business internal active structure elements that work together to perform collective behaviour.
- **Business interface:** a point of access where a business service is made available to the environment.
- **Business object:** a concept used within a particular business domain.
- **Representation:** the perceptible form of the information carried by a business object.

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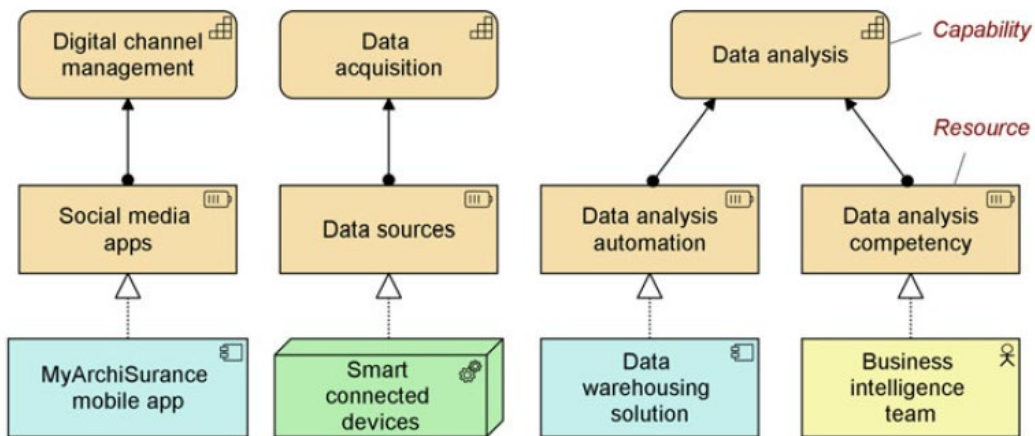


Fig. 5.8 Resources assigned to capabilities and realised by structure elements

Strategy Concepts

- **Course of action:** an approach or plan for configuring some capabilities and resources of the enterprise, undertaken to achieve a goal.
- **Capability:** an ability that an active structure element, such as an organization, person or system, possesses.
- **Resource:** an asset owned or controlled by an individual or organisation.

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Business Layer Concepts

- **Business Structure Concepts**
- **Business Behaviour Concepts**
- **Higher-Level Business Concepts**

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An example of a business layer model is shown in Fig. 5.9, illustrating the use of the central concepts and relations. In the following subsections, we will explain the business layer concepts in more detail.

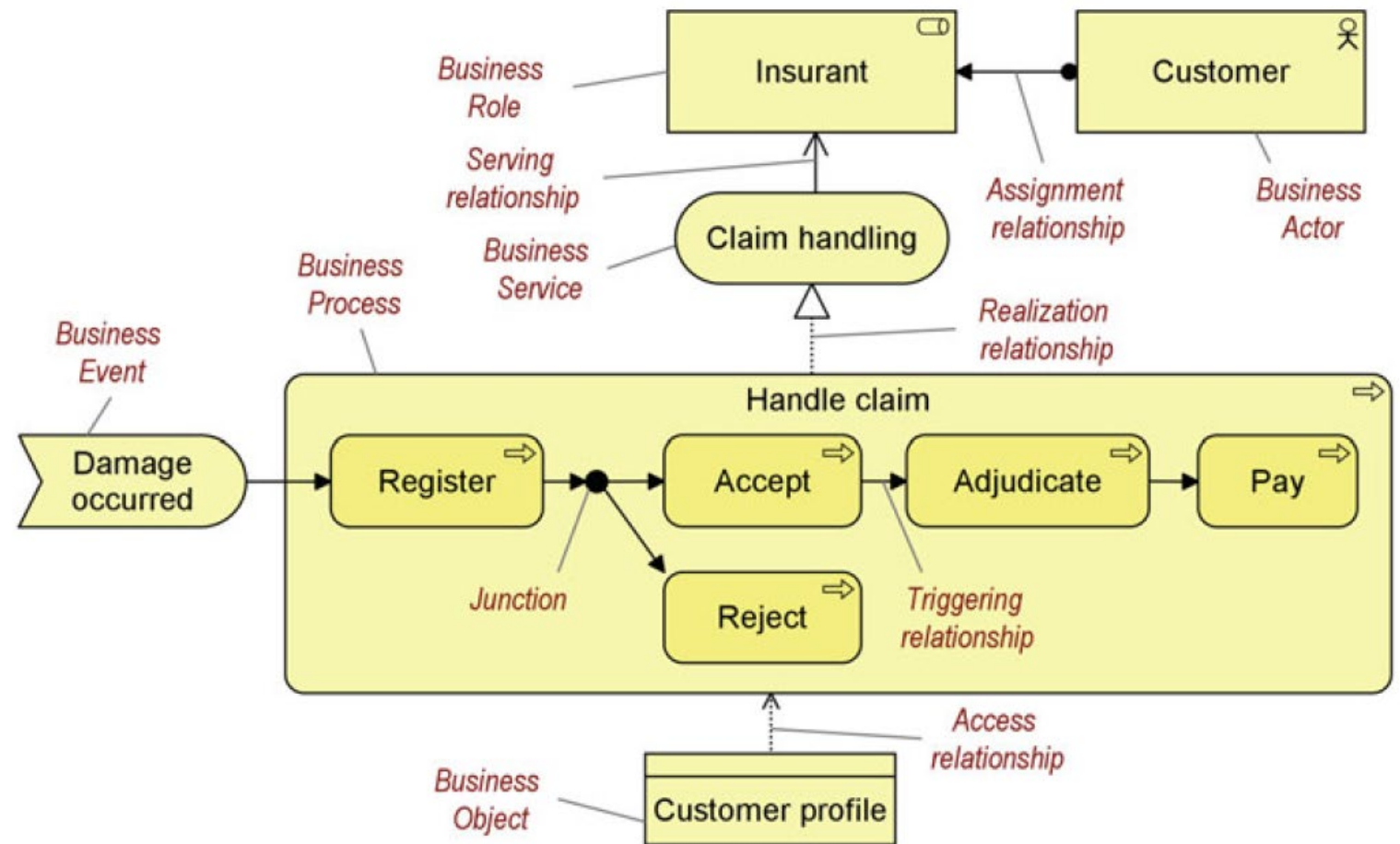


Fig. 5.9 Example of a business layer model

A LANGUAGE FOR ENTERPRISE MODELLING

Business Behavior Concepts

- **Business service:** an explicitly defined exposed business behaviour.
- **Business process:** a sequence of business behaviours that achieves a specific outcome such as a defined set of products and services.
- **Business function:** a collection of business behaviour based on a chosen set of criteria (typically required business resources and/or competences), closely aligned to an organisation, but not necessarily explicitly governed by the organisation.
- **Business interaction:** a unit of collective business behaviour performed by (a collaboration of) two or more business roles.
- **Business event:** a business behaviour element that denotes an organizational state change. It may originate from and be resolved inside or outside the organisation.

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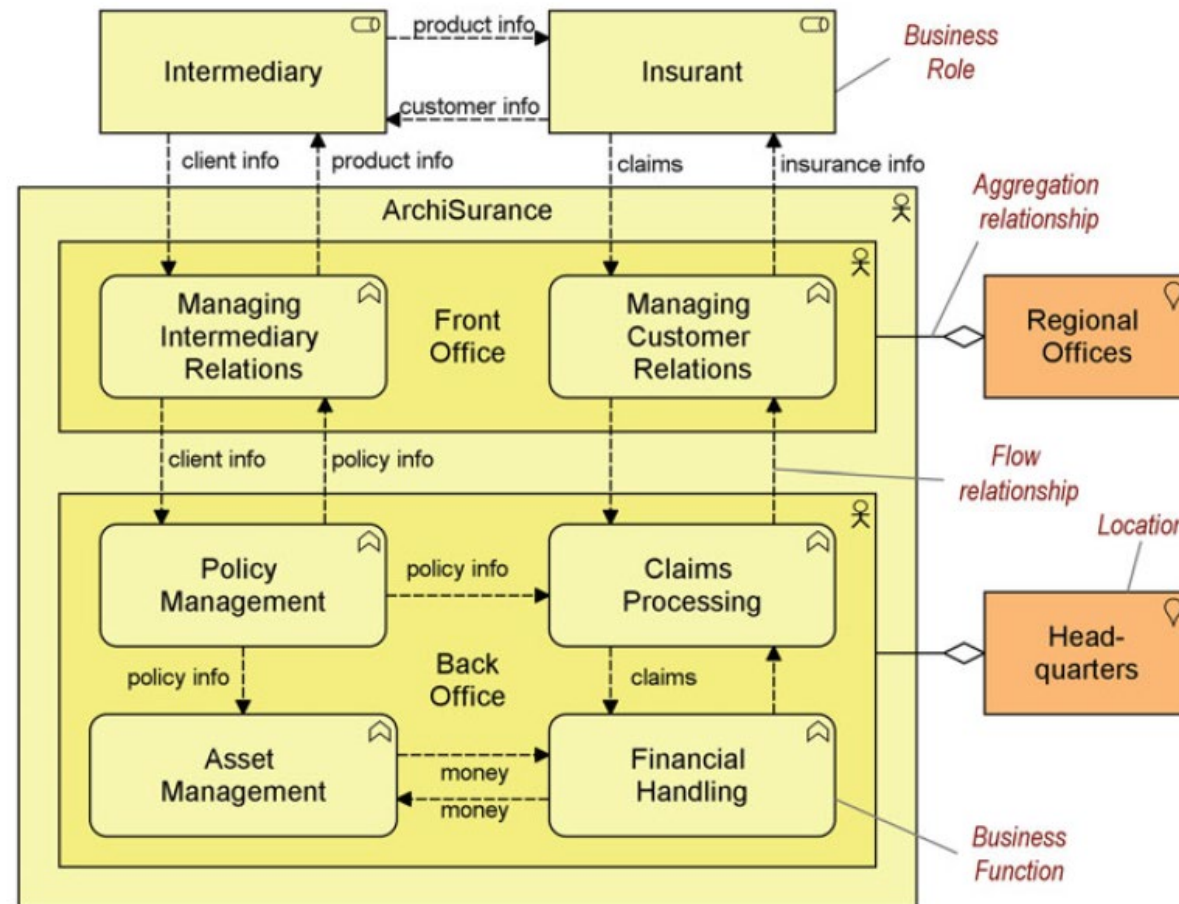


Fig. 5.10 Business functions, organisation structure and location of ArchiSurance

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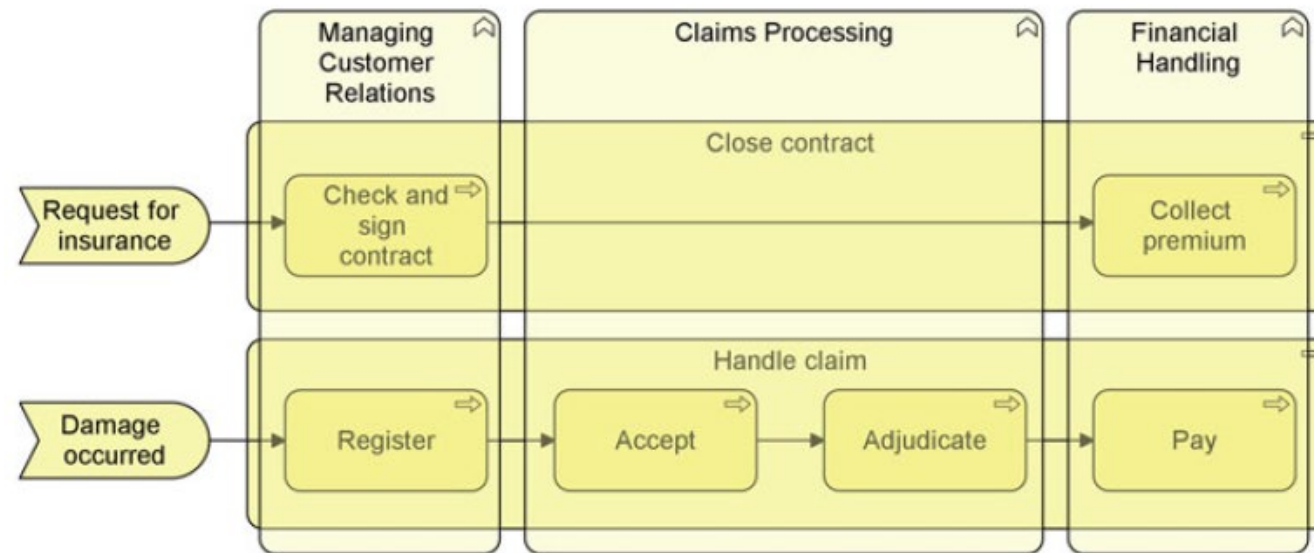
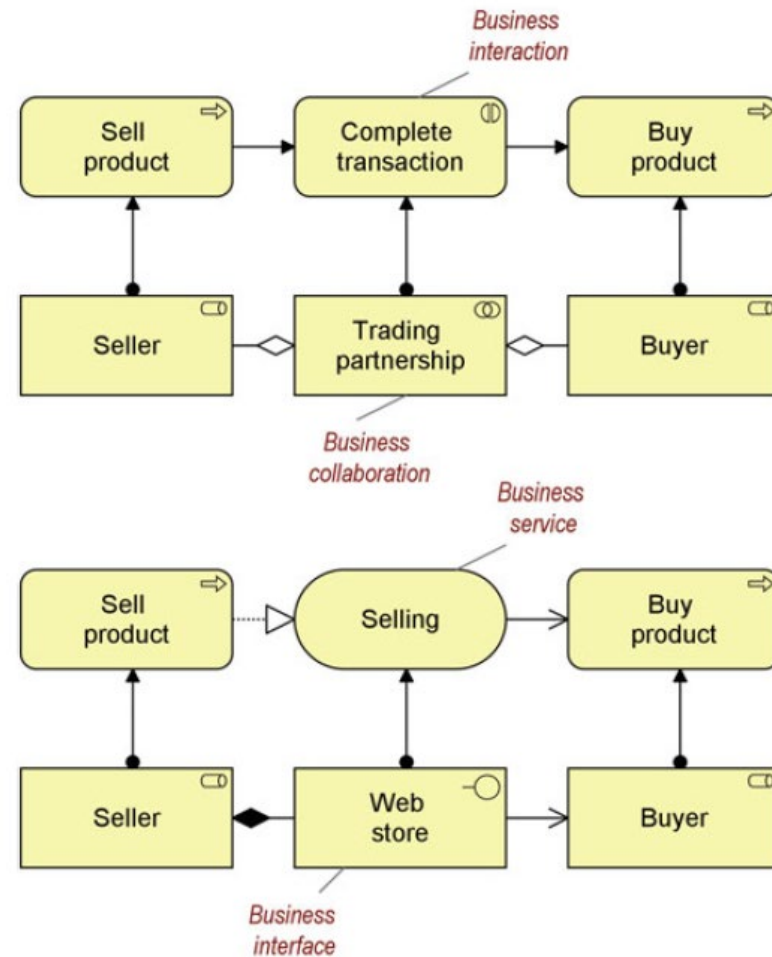


Fig. 5.11 Business processes versus business functions

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Fig. 5.12 Interaction versus service use



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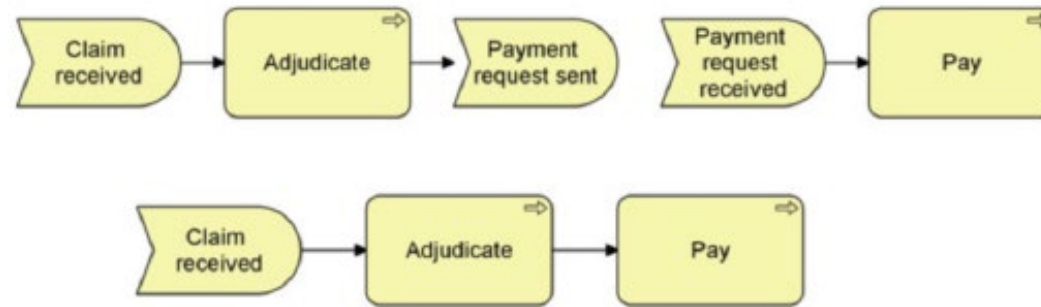


Fig. 5.13 Events to decouple processes

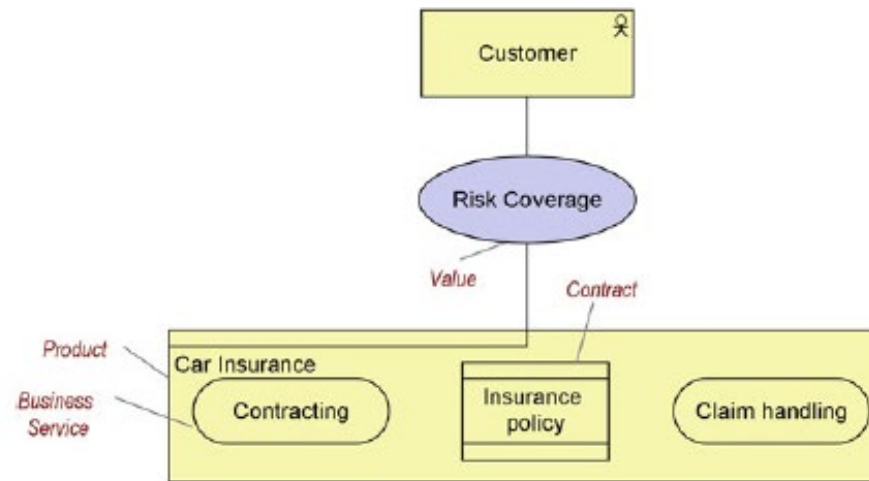


Fig. 5.14 Services grouped into a product

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Higher-Level Business Concepts

- **Product:** a coherent collection of services and/or passive structure elements, accompanied by a contract/set of agreements, which is offered as a whole to (internal or external) customers.
- **Contract:** a formal or informal specification of an agreement between a provider and a consumer that specifies the rights and obligations associated with a product and establishes functional and non-functional parameters for interaction.

A LANGUAGE FOR ENTERPRISE MODELLING

Application Layer Concepts

- **Application Structure Concepts**
- **Application Behavior Concepts**

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A typical example of an application layer model is shown in Fig. 5.15, illustrating the use of the central concepts.

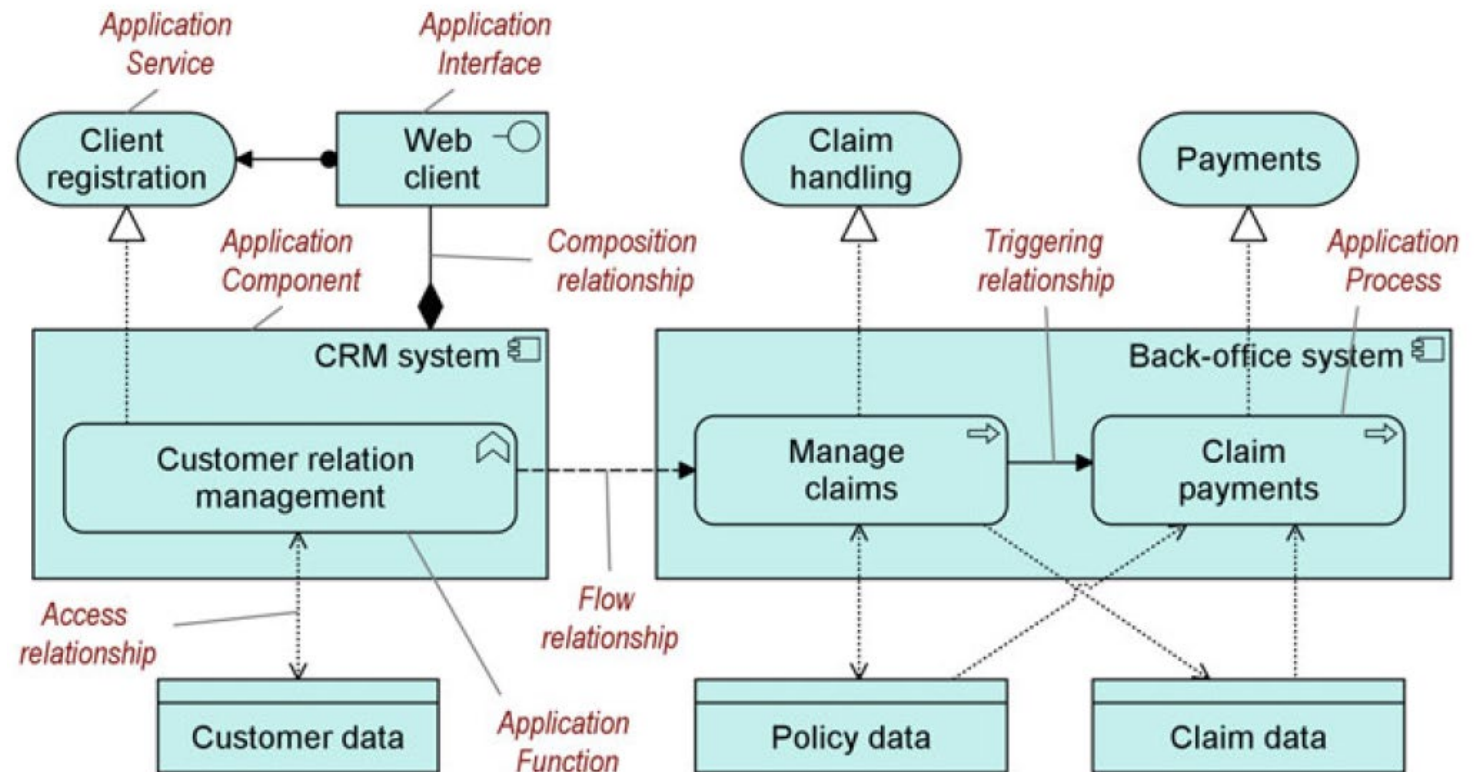


Fig. 5.15 Example of an application layer model

A LANGUAGE FOR ENTERPRISE MODELLING

Application Structure Concepts

- **Application component:** an encapsulation of application functionality aligned to implementation structure, which is modular and replaceable. It encapsulates its behaviour and data, exposes services and makes them available through interfaces.
- **Application collaboration:** an aggregate of two or more application components that work together to perform collective application behaviour.
- **Application interface:** a point of access where an application service is made available to a user, another application component or a node.
- **Data object:** data structured for automated processing.

A LANGUAGE FOR ENTERPRISE MODELLING

Application Behaviour Concepts

- **Application service:** an explicitly defined exposed application behaviour.
- **Application function:** automated behaviour that can be performed by an application component.
- **Application process:** a sequence of application behaviours that achieves a specific outcome.
- **Application interaction:** a unit of collective application behaviour performed by (a collaboration of) two or more application components.
- **Application event:** an application behaviour element that denotes a state change.

A LANGUAGE FOR ENTERPRISE MODELLING

The application layer and the business layer can be easily linked in ArchiMate (Fig. 5.16). Two types of relations provide this link:

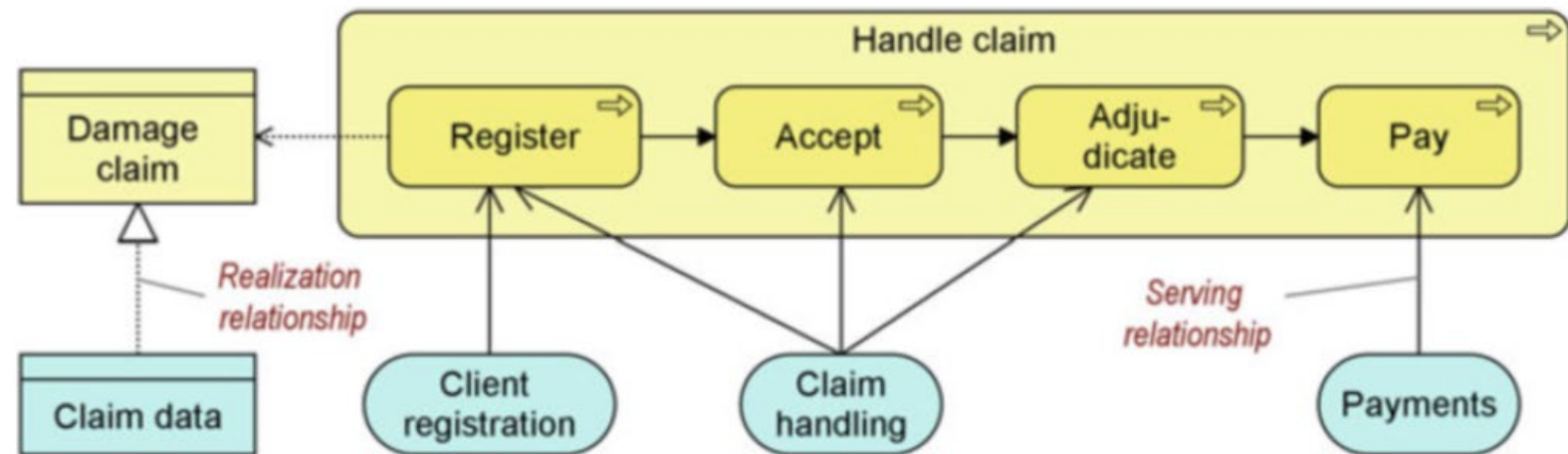


Fig. 5.16 Example of a business-application alignment model

A LANGUAGE FOR ENTERPRISE MODELLING

Technology Layer Concepts

- Technology Structure Concepts
- Technology Behavior Concepts

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A typical example of a technology layer model is shown in Fig. 5.17, illustrating the use of the central concepts.

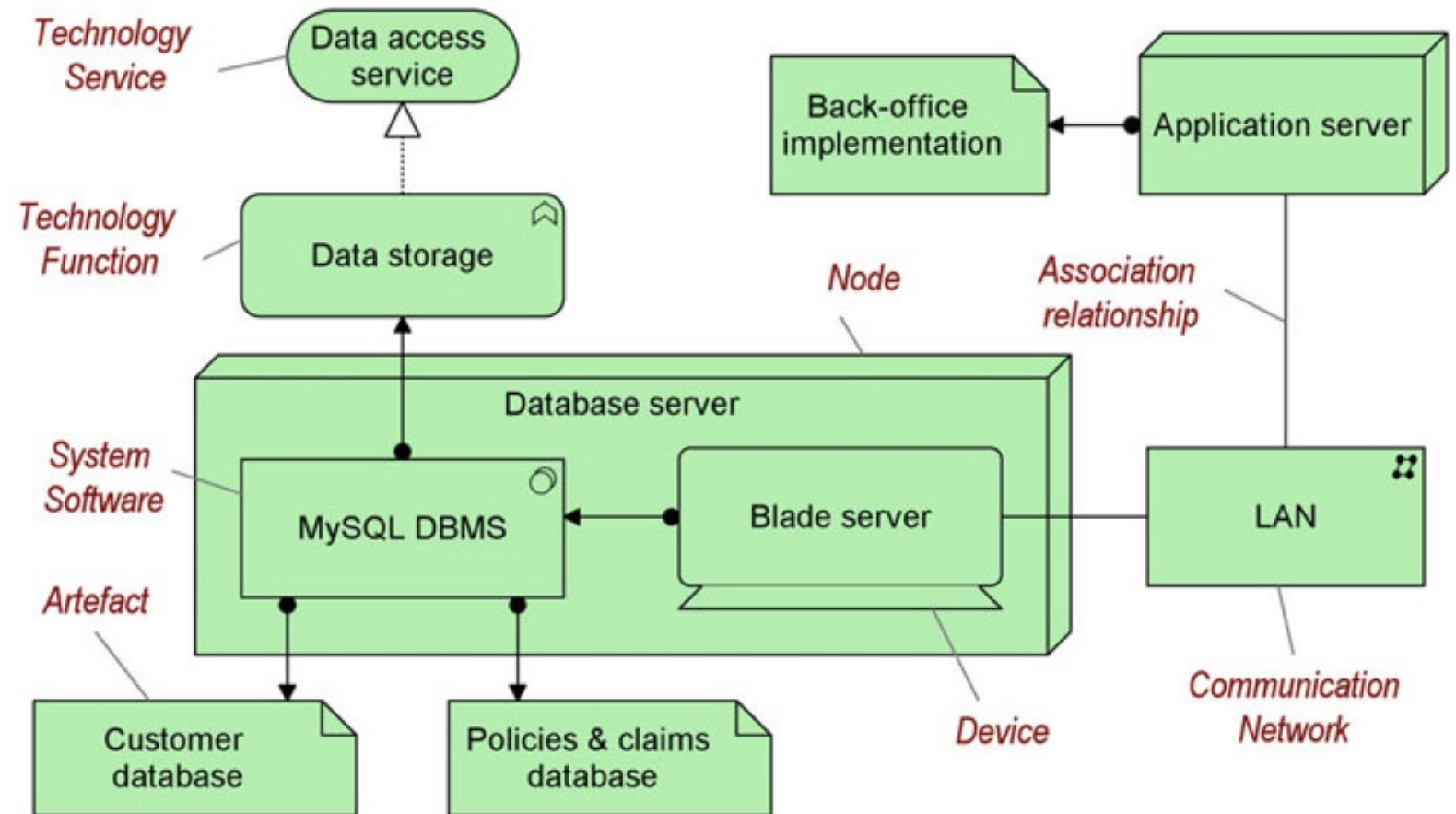


Fig. 5.17 Example of a technology layer model

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Technology Structure Concepts

- **Node:** a computational or physical resource that hosts, manipulates or interacts with other computational or physical resources.
- **Technology collaboration:** an aggregate of two or more nodes that work together to perform collective technology behaviour.
- **Technology interface:** a point of access where technology services offered by a node can be accessed.
- **Device:** a physical IT resource upon which system software and artefacts may be deployed for execution.
- **System software:** software that provides or contributes to an environment for storing, executing and using software or data deployed within it.
- **Path:** a link between two or more nodes, through which these nodes can exchange data or material.
- **Communication network:** a set of structures that connects computer systems or other electronic devices for transmission, routing and reception of data or data-based communications such as voice and video.
- **Artefact:** a piece of data that is used or produced in a software development process or by deployment and operation of a system.

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Technology Behaviour Concepts

- **Technology service:** externally visible unit of functionality, provided by one or more nodes, exposed through well-defined interfaces, and meaningful to the environment.
- **Technology function:** a behaviour element that groups infrastructural behaviour that can be performed by a node.
- **Technology process:** a sequence of technology behaviours that achieves a specific outcome.
- **Technology interaction:** a unit of collective technology behaviour performed by (a collaboration of) two or more nodes.
- **Technology event:** a technology behaviour element that denotes a state change.

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Application–Technology Alignment

The technology layer and the application layer can also be linked very easily. Similar to business–application alignment, two types of relations provide this link:

- 1) Technology services can serve application functions and technology interface serve application components, i.e. there is a support relation between the technology and application layers (Fig. 5.18).
- 2) Artefacts can realize data objects and application components, i.e. there is an implementation relation between the technology and application layers (Fig. 5.18).

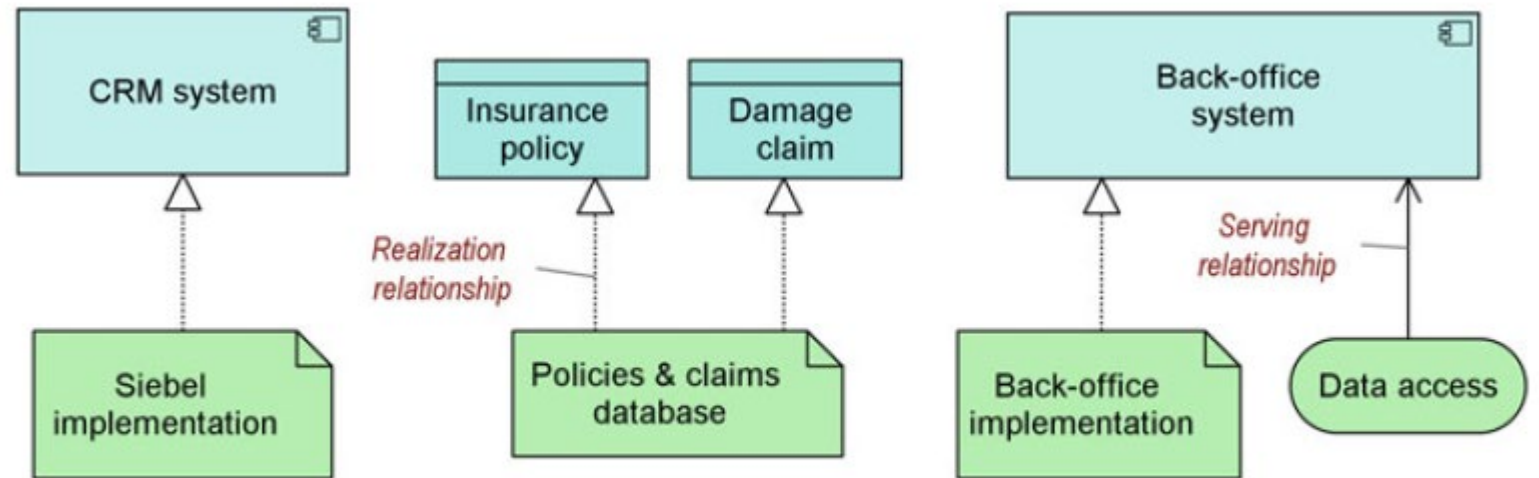


Fig. 5.18 Example of applications and data supported by technology

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Physical Concepts

- **Equipment:** one or more physical machines, tools or instruments that can create, use, store, move or transform materials.
- **Facility:** a physical structure or environment.
- **Distribution network:** a physical network used to transport materials or energy.
- **Material:** tangible physical matter or physical elements.

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Physical Concepts

Modelling the physical world and its interplay with IT is therefore an important addition to the language in ArchiMate version 3.0 (The Open Group 2016a). An example of these concepts is shown in Fig. 5.19.

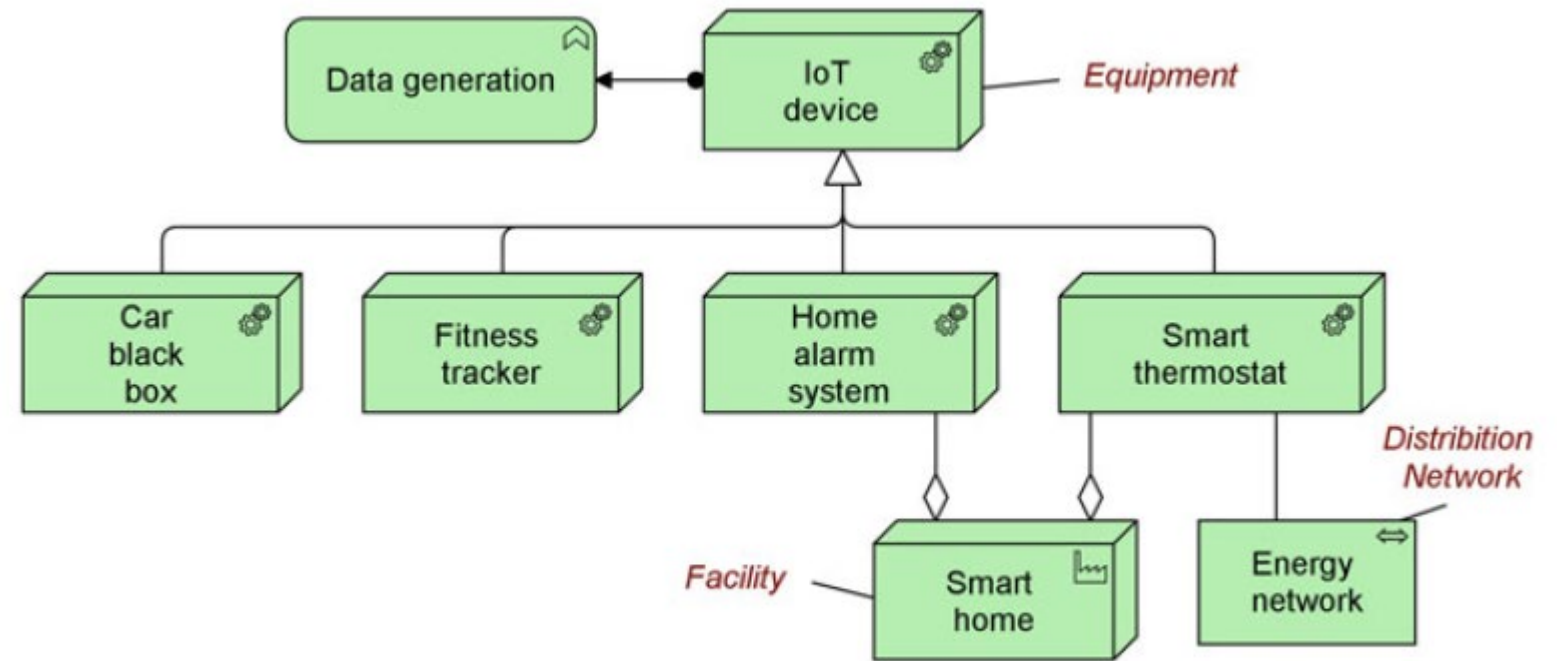


Fig. 5.19 Example of physical concepts

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Implementation and Migration Concepts

- Implementation-Related Concepts
- Migration Planning Concepts

A LANGUAGE FOR ENTERPRISE MODELLING

Implementation-Related Concepts

- **Work package:** a series of actions identified and designed to achieve specific results within specified time and resource constraints.
- **Implementation event:** a behaviour element that denotes a state change related to implementation or migration.
- **Deliverable:** a precisely defined outcome of a work package.

A LANGUAGE FOR ENTERPRISE MODELLING

Implementation-Related Concepts

An example of a model using the implementation-related concepts is shown in Fig. 5.20, illustrating the use of the central concepts and relations for expressing the work that needs to be done to realise an architecture.

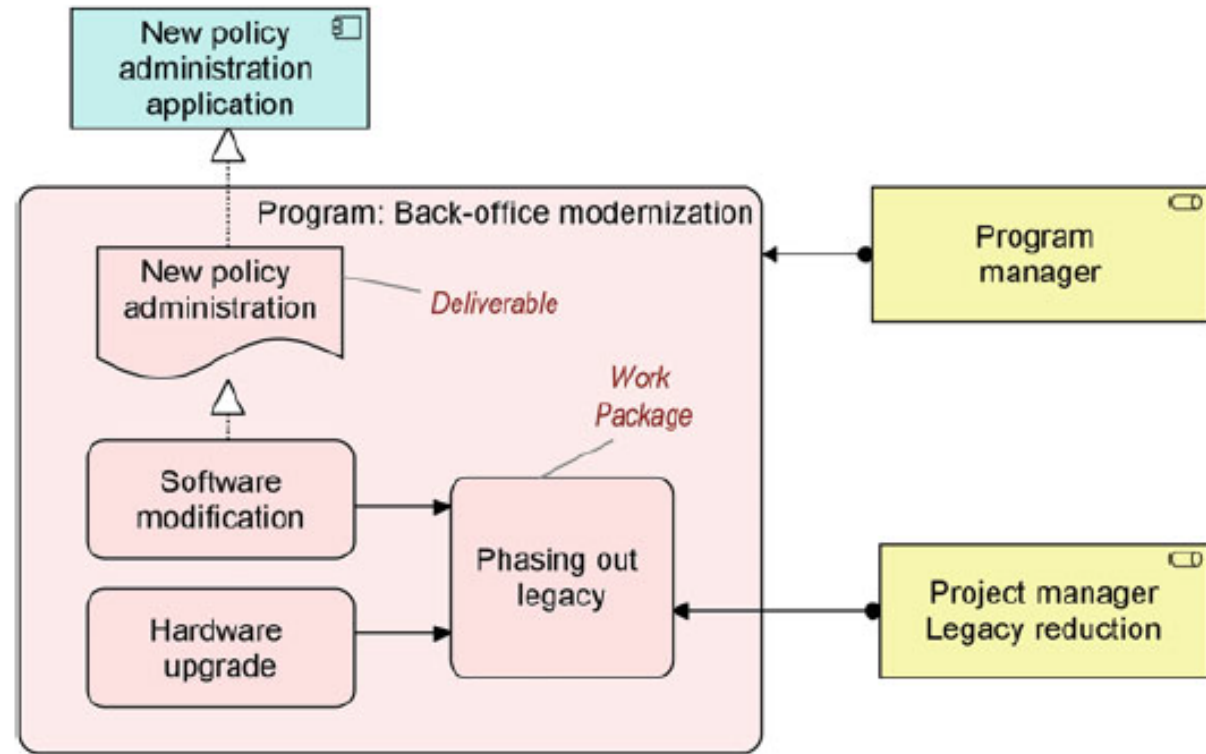


Fig. 5.20 Example of an implementation model

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Migration Planning Concepts

- **Plateau:** a relatively stable state of the architecture that exists during a limited period of time.
- **Gap:** a statement of difference between two plateaus.

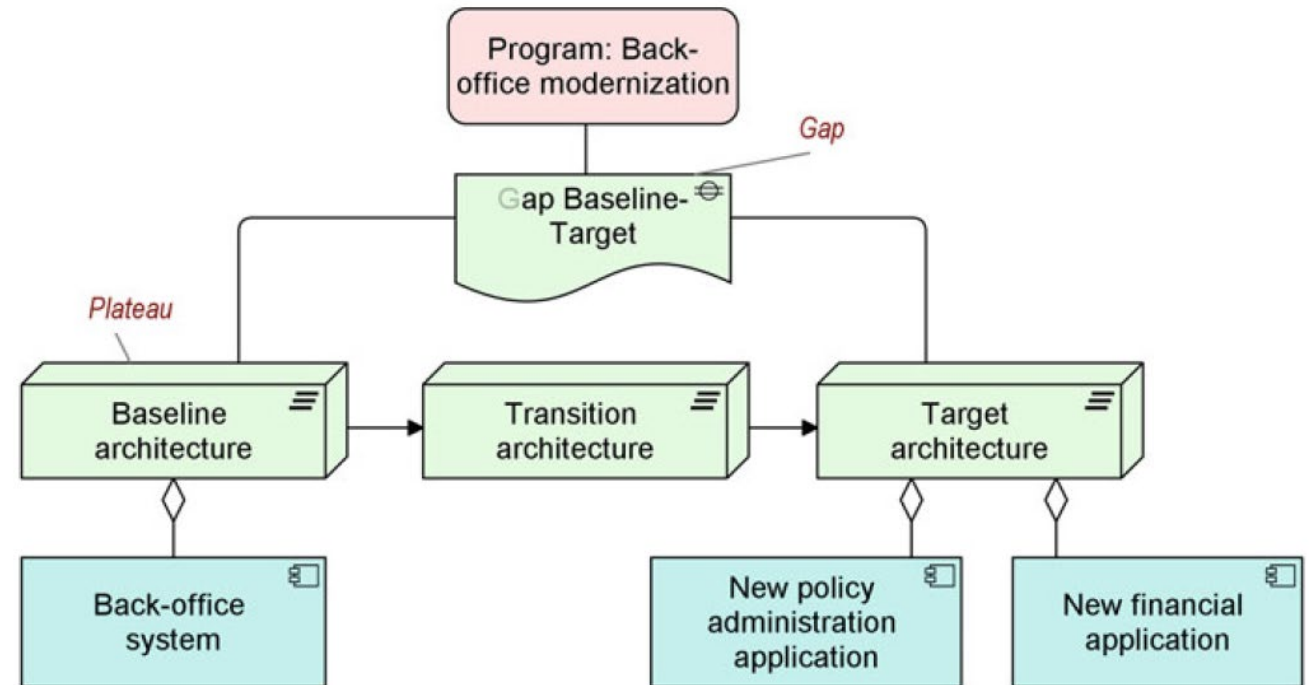


Fig. 5.21 Example of a migration model

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RELATIONS

Table 5.1 Structural and dependency relations

Association —————	The <i>association</i> models a relation between objects that is not covered by another, more specific relationship
Influence +/- ----->	The <i>influence</i> relation models that an element affects the implementation or achievement of some motivation element
Access>	The <i>access</i> relation models that behavioural elements can observe or act upon passive structure elements
Serving —————>	The <i>serving</i> relation models that an element offers its functionality to another element
Realisation -----▷	The <i>realisation</i> relation indicates that an entity plays a critical role in the creation, achievement, sustenance, or operation of a more abstract entity
Specialisation —————▷	The <i>specialisation</i> relation indicates that an element is a particular kind of another element
Assignment ●————→	The <i>assignment</i> relation expresses the allocation of responsibility, performance of behaviour or execution
Aggregation ◇—————	The <i>aggregation</i> relation indicates that an element groups a number of other elements
Composition ◆—————	The <i>composition</i> relation indicates that an element consists of a number of other elements

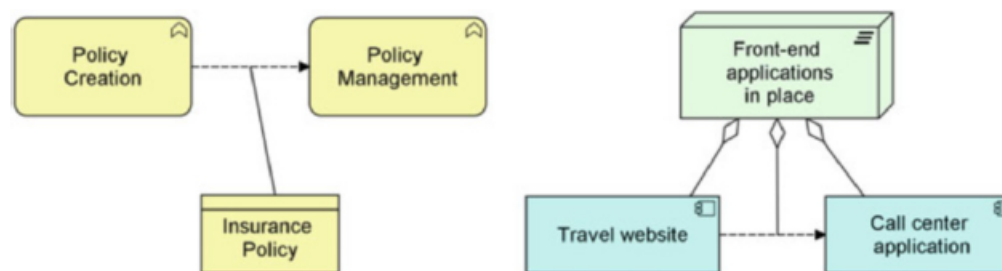


Fig. 5.22 Relationships to relationships

A LANGUAGE FOR ENTERPRISE MODELLING

Guidelines for Modelling

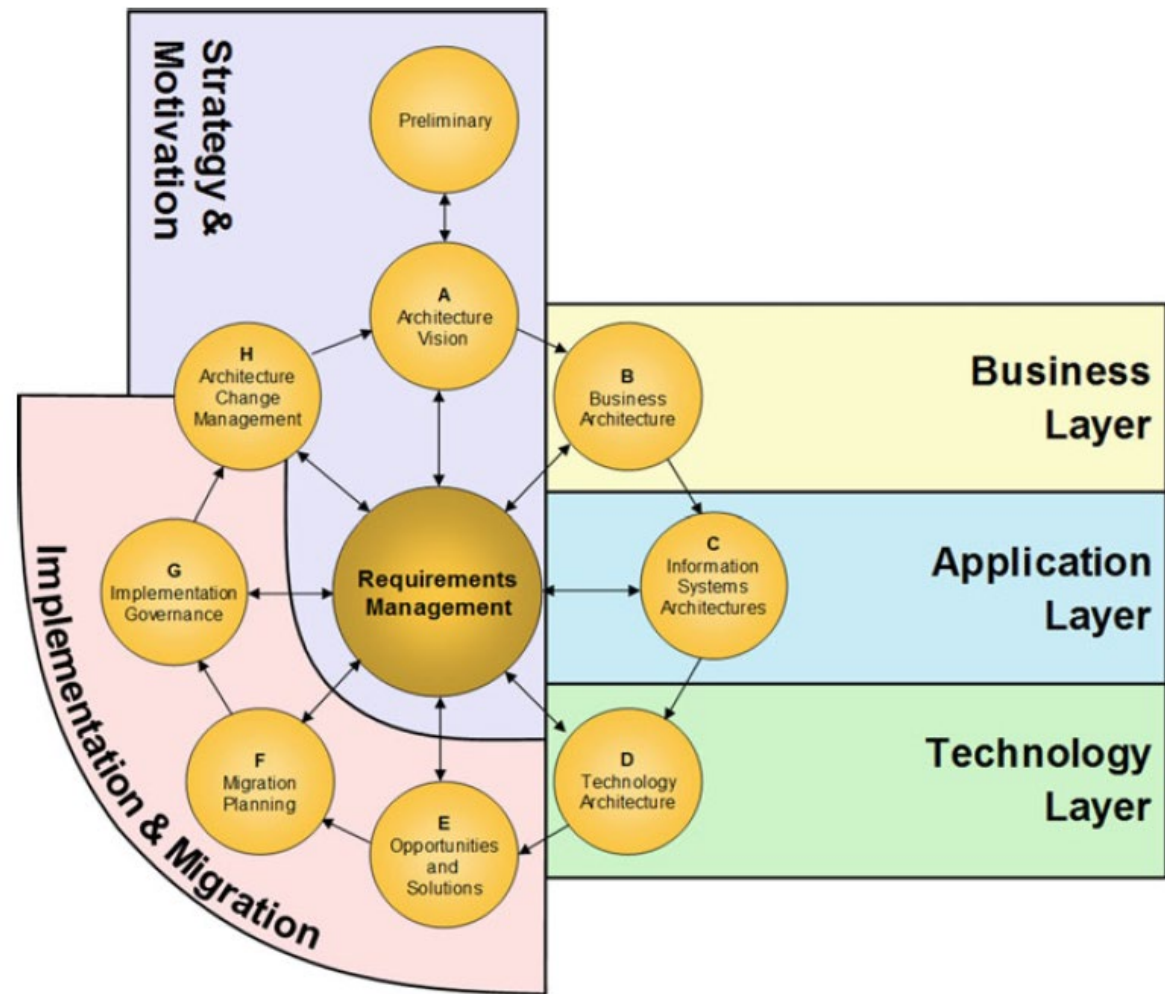


Fig. 6.14 Correspondence between TOGAF ADM and ArchiMate language

A LANGUAGE FOR ENTERPRISE MODELLING

Discussion

According to the below paper describe **how the Business Model Canvas can mapped to ArchiMate?**

- *Meertens, L. O., Iacob, M. E., Nieuwenhuis, L. J., van Sinderen, M. J., Jonkers, H., & Quartel, D. (2012, March). Mapping the business model canvas to ArchiMate. In Proceedings of the 27th annual ACM symposium on applied computing (pp. 1694-1701).*



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- Kotusev, S., & Kurnia, S. (2021). The theoretical basis of enterprise architecture: A critical review and taxonomy of relevant theories. *Journal of Information Technology*, 36(3), 275-315.
- Meertens, L. O., Iacob, M. E., Nieuwenhuis, L. J., van Sinderen, M. J., Jonkers, H., & Quartel, D. (2012, March). Mapping the business model canvas to ArchiMate. In *Proceedings of the 27th annual ACM symposium on applied computing* (pp. 1694-1701).

